Jessica U. Ezemba

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Ph.D., Mechanical Engineering | GPA: 3.92/4.0 — Expected May 2027
M.S., Integrated Innovation for Products & Services | GPA: 3.96/4.0 — Dec. 2022

Texas A&M University — College Station, TX
B.S., Mechanical Engineering | GPA: 3.52/4.0 — May 2021

SKILLS

Programming & Engineering: Python, FEniCSx, PyAnsys, ANSYS, MATLAB, SOLIDWORKS, CATIA, AutoCAD, PTC Creo
AI/ML: PyTorch, PyTorch Geometric, Multi-Agent Systems, LangGraph, Vision Transformers, GNNs, Geometric Deep Learning, Autoencoders, LLMs, RAG, MCP
Computation & DevOps: HPC, FEA, CFD, Probabilistic Simulation, Surrogate Modeling, Neural Operators

RESEARCH EXPERIENCE

Carnegie Mellon University — Pittsburgh, PA
PhD Candidate, Mechanical Engineering — Aug. 2023 - Present

- Built custom Multi-Agent System (MAS) for engineering design and simulation utilizing organizational design principles, increasing computational efficiency by 50%
- Developed Model Context Protocol (MCP) server for prompt-to-CAD, parametric geometry variation, and FEA/CFD simulations, enabling faster integration into MAS and reducing computational costs by 50%
- Created comprehensive benchmark datasets for evaluating LLM performance on engineering simulation reasoning across text, image, and video modalities, establishing new academic standards
- Engineered neural network surrogate models with FEniCSx to quantify uncertainty in complex engineering simulations, reducing computational costs by 90%
- Advanced the application of Geometric Deep Learning to Graph Neural Networks, creating custom data loaders that effectively process complex 3D unstructured mesh geometries
- Integrated state-of-the-art LLMs and Vision Transformers for multimodal engineering data interpretation

PROFESSIONAL EXPERIENCE

KLA — Ann Arbor, MI
Advanced Mechatronics Intern — May 2022 - Aug. 2022 & May 2023 - Aug. 2023

- Developed a 3D parametric geometry and physics-based optimizer to reduce the time to design flexures
- Created a 3D design exploration algorithm using ANSYS for fatigue, buckling, and frequency analysis
- Developed an optimization algorithm in MATLAB to decrease controller tuning time by over 50%

Symbotic — Wilmington, MA
Systems Design Intern — Jan. 2023 - May 2023

- Increased warehouse automation throughput using FEA to optimize component mass and structural integrity of multiple 3D component parts
- Designed high-risk safety devices for warehouse staff, achieving 100% compliance with safety regulations
- Developed integrated documentation covering multiple subsystem aspects, improving overall system understanding

SELECTED PUBLICATIONS

1. **Ezemba, J.,** et al. (2026). "Organizationally-Intelligent Agents: Connecting Multi-Agent Engineering Systems to Organizational Design Theory." Intl. Conf. on Design Computing and Cognition. (In Review)
2. Mehta, M.D.*, **Ezemba, J.*,** et al. (2026). "VizArQA: A Benchmark for Visual Question Answering in Architectural Simulation." J. of Eng. for Sustainable Buildings and Cities. (In Review)
3. **Ezemba, J.,** Pohl, J., Tucker, C., & McComb, C. (2025). "OpenSeeSimE: A Large-Scale Benchmark to Assess Vision-Language Model QA Capabilities in Engineering Simulations." (In Review)
4. **Ezemba, J.,** et al. (2025). "Simulation vs. Hallucination: Assessing VLM QA Capabilities in Engineering Simulations." Proc. 7th Workshop on Design Automation for CPS and IoT, pp. 1-9.
5. **Ezemba, J.,** McComb, C., & Tucker, C. (2025). "Neural Network Surrogate Modeling for Stochastic FEM using 3D Graph Representations." J. of Mechanical Design, pp. 1-15.